

AutF2

Computations for the automorphisms group of F_2

2.0

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Óscar Fernández Ayala

Óscar Fernández Ayala Email: oscar.fernandez-ayala@tu-braunschweig.de
Homepage: <https://osferay.github.io/>

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Chapter 1

Preface

In this package we provide methods to compute with automorphisms of the free group of rank 2. The details of the algorithms are available in [Fer26]. We use the following notation:

$$\sigma = \{x \mapsto y, y \mapsto x, \quad \phi_1 = \{x \mapsto xy^{-1}, y \mapsto y, \quad \phi_2 = \{x \mapsto x, y \mapsto yx, \quad \phi_3 = \{x \mapsto y^{-1}x, y \mapsto y.$$

Then $\text{Aut}(F_2)$ has the following presentation.

$$\begin{aligned} \text{Aut}(F_2) = \langle \sigma, \phi_1, \phi_2, \phi_3 \mid & \phi_1 \phi_3 = \phi_3 \phi_1, \quad \phi_2 \phi_3 \phi_2 = \phi_3 \phi_2 \phi_3, \quad \phi_2 \phi_1 \phi_2 = \phi_1 \phi_2 \phi_1, \\ & (\phi_1 \phi_2 \phi_3)^4 = 1, \quad \sigma^2 = 1, \quad \phi_1^\sigma = \phi_2^{-1}, \\ & \phi_3 \sigma = \sigma \phi_3^{-1} \phi_2^{-1} \phi_1^{-1} \phi_2 \phi_3, \quad \phi_3^{-1} \sigma = \sigma \phi_3^{-1} \phi_2^{-1} \phi_1 \phi_2 \phi_3 \rangle. \end{aligned}$$

Denote the commutator subgroup of a group G by G' . For $\psi \in \text{Aut}(F_2)$, we define the map ψ^* by $\psi^*(xF'_2) = \psi(x)F'_2$. If $\bar{\cdot}$ denotes the isomorphism between $\text{Aut}(F_2/F'_2)$ and $\text{GL}_2(\mathbb{Z})$, then the map

$$\begin{aligned} \Psi: \text{Aut}(F_2) &\rightarrow \text{GL}_2(\mathbb{Z}) \\ \psi &\mapsto \overline{(\psi^*)} \end{aligned}$$

is a group homomorphism and the kernel of this map is $\text{Inn}(F_2)$. We denote by SA_2 the subgroup of $\text{Aut}(F_2)$ consisting of those automorphisms that are mapped into $\text{SL}_2(\mathbb{Z})$ under Ψ . Let B_4 denote the braid group on 4 strands. Then $\text{SA}_2 \cong B_4/Z(B_4)$. Using this, we have that each automorphism can be written uniquely as a word of the form

$$\alpha = \sigma^{\varepsilon_1} \Delta^{\varepsilon_2} \alpha_1 \alpha_2 \cdots \alpha_k$$

where $\varepsilon_1, \varepsilon_2 \in \{0, 1\}$, $\Delta = \phi_1(\phi_2 \phi_1)(\phi_3 \phi_2 \phi_1)$, and $\alpha_1, \dots, \alpha_k \in \text{SA}_2$ whose images under the isomorphism between SA_2 and $B_4/Z(B_4)$ are simple braids. We call this expression the left canonical form of α .

Chapter 2

Automorphisms

2.1 Automorphisms

2.1.1 IsAutomorphismOffF2 (for IsObject)

▷ IsAutomorphismOffF2(*arg*) (Category)
Returns: true or false

2.1.2 AutomorphismOffF2 (for IsFreeGroup, IsList)

▷ AutomorphismOffF2(*F*, *list*) (operation)

Constructor of the automorphism object given a word of automorphisms.

2.1.3 WordOfAutomorphismOffF2 (for IsAutomorphismOffF2)

▷ WordOfAutomorphismOffF2(*aut*) (attribute)

Returns the left canonical form of the given automorphism.

2.1.4 ImagesAutomorphismOffF2 (for IsAutomorphismOffF2)

▷ ImagesAutomorphismOffF2(*aut*) (attribute)

Returns the images of the generators of F_2 under the given automorphism.

2.1.5 ImageByAutomorphismOffF2 (for IsAutomorphismOffF2, IsAssocWordWithInverse)

▷ ImageByAutomorphismOffF2(*aut*, *w*) (operation)

Returns the image of the given word under the given automorphism.

2.1.6 IsIdentityAutomorphismOfF2 (for IsAutomorphismOfF2)

- ▷ IsIdentityAutomorphismOfF2(*aut*) (property)
Returns: true or false

2.1.7 MatrixRepresentationOfAutomorphismOfF2 (for IsAutomorphismOfF2)

- ▷ MatrixRepresentationOfAutomorphismOfF2(*aut*) (attribute)

Returns the image of an automorphism in $\text{GL}_2(\mathbb{Z})$ under the map Ψ .

2.1.8 IsSpecialAutomorphismOfF2 (for IsAutomorphismOfF2)

- ▷ IsSpecialAutomorphismOfF2(*aut*) (property)
Returns: true or false

2.1.9 IsConjugacyAutomorphismOfF2 (for IsAutomorphismOfF2)

- ▷ IsConjugacyAutomorphismOfF2(*aut*) (property)
Returns: true or false

2.1.10 ConjugacyElementConjugacyAutomorphismOfF2 (for IsAutomorphismOfF2)

- ▷ ConjugacyElementConjugacyAutomorphismOfF2(*aut*) (attribute)

Getter of the element that determines a conjugacy automorphism.

2.1.11 ConjugacyAutomorphismOfF2 (for IsFreeGroup, IsAssocWordWithInverse)

- ▷ ConjugacyAutomorphismOfF2(*F*, *w*) (operation)

Returns the automorphism determined by conjugacy of the given word in F_2

2.1.12 AutomorphismOfF2ByImages (for IsFreeGroup, IsAssocWordWithInverse, IsAssocWordWithInverse)

- ▷ AutomorphismOfF2ByImages(*F*, *u*, *v*) (operation)

Returns an automorphism such that the images of the basis elements are the given words *u* and *v*. If the automorphism does not exist, it returns false.

2.1.13 AutomorphismOfF2ByMatrix (for IsFreeGroup, IsList)

- ▷ AutomorphismOfF2ByMatrix(*F*, *M*) (operation)

Returns an automorphism whose image in $\text{GL}_2(\mathbb{Z})$ is the given matrix.

Chapter 3

Conjugacy, fixed-point subgroups, and centralizers

3.1 Centralizer

3.1.1 CentralizerAutomorphismOfF2

▷ `CentralizerAutomorphismOfF2(aut)` (function)

Returns a generating set for the centralizer of a given automorphism of F_2 .

3.2 Conjugacy

3.2.1 AreConjugateAutomorphismsOfF2

▷ `AreConjugateAutomorphismsOfF2(a, b)` (function)

Returns a conjugating element if the automorphisms a and b are conjugate. Otherwise, it returns `false`.

3.3 Fixed points subgroup

3.3.1 FixedSubgroupAutomorphismOfF2

▷ `FixedSubgroupAutomorphismOfF2(aut)` (function)

Returns a basis for the fixed-point subgroup of a given automorphism.

Chapter 4

Related Algorithms

4.1 Matrix groups

4.1.1 ConjugacyGL2Z

▷ `ConjugacyGL2Z(A, B)` (function)

Decides whether two 2×2 matrices in $\text{GL}_2(\mathbb{Z})$ are conjugate. If so, returns the element C such that $A = C^{-1}BC$.

4.1.2 CentralizerGL2Z

▷ `CentralizerGL2Z(A, B)` (function)

Given a matrix A , computes a matrix C and an integer n such that $\text{Cent}_{\text{GL}_2(\mathbb{Z})}(A) = \langle -C, C \rangle$ and $C^n = A$.

4.1.3 MembershipSubgroupSL2Z

▷ `MembershipSubgroupSL2Z(gens, A)` (function)

Given a matrix A , decides whether $A \in \langle \text{gens} \rangle$ and, if so, computes a word w in the generators representing A .

4.2 Free groups

4.2.1 AreAutomorphicEquivalent (for IsFreeGroup, IsAssocWordWithInverse, IsAssocWordWithInverse)

▷ `AreAutomorphicEquivalent(F, u, v)` (operation)

Decides whether u and v are automorphically equivalent in F .

4.2.2 IsCyclicallyReducedWord (for IsAssocWordWithInverse)

▷ IsCyclicallyReducedWord(u) (property)
Returns: true or false

4.2.3 IsSubword (for IsAssocWordWithInverse, IsAssocWordWithInverse)

▷ IsSubword(u, v) (operation)

Returns whether the word v is a subword of u or not.

4.2.4 IsPrefix (for IsAssocWordWithInverse, IsAssocWordWithInverse)

▷ IsPrefix(u, v) (operation)

Returns whether the word v is a prefix of u .

4.2.5 IsSuffix (for IsAssocWordWithInverse, IsAssocWordWithInverse)

▷ IsSuffix(u, v) (operation)

Returns whether the word v is a suffix of u .

4.2.6 RootFreeGroup (for IsAssocWordWithInverse)

▷ RootFreeGroup(v) (operation)

Returns a word w and an integer r such that $w^r = v$ and r is maximal.

4.2.7 NielsenReducedSet

▷ NielsenReducedSet(V) (function)

Given a sequence of elements V of a free group, returns a Nielsen reduced set W that is equivalent to V .

4.2.8 IsNielsenReducedSet

▷ IsNielsenReducedSet(V) (function)

Given a sequence of elements V of a free group, returns whether V is a Nielsen reduced set.

4.2.9 InFreeSubgroup

▷ InFreeSubgroup(V, w) (function)

Given a sequence of elements V of a free group and a word w , returns whether w is in the subgroup generated by V .

4.2.10 IsSubgroupOfFreeSubgroup

▷ `IsSubgroupOfFreeSubgroup( $V$ ,  $W$ )` (function)

Given two sequences of elements V, W of a free group, returns whether the subgroup generated by W is a subgroup of the group generated by V .

4.2.11 IsNormalSubgroupOfFreeSubgroup

▷ `IsNormalSubgroupOfFreeSubgroup( $V$ ,  $W$ )` (function)

Given two sequences of elements V, W of a free group, returns whether the subgroup generated by W is a normal subgroup of the group generated by V .

4.2.12 AreEquivalent

▷ `AreEquivalent( $V$ ,  $W$ )` (function)

Given two sequences of elements V, W of a free group, returns whether V and W are equivalent.

References

[Fer26] Ó. Fernández Ayala. Practical computations with automorphisms of f_2 . *Submitted*, 2026. [3](#)

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